llmXive Automated Review of EvalVerse: Pipeline-Aware and Expert-Calibrated Benchmarking for Professional Cinematic Video Generation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper generally maintains strong alignment between its claims and citations, particularly regarding the taxonomy and methodology. However, there are specific instances where textual claims regarding model versions and "firsts" require tighter verification against the bibliography and comparative tables.

First, in Section 5 ("Dataset Curation: Test Pair Construction"), the authors state: "We utilize a Gemini 3.1 Pro~\cite{gemini}... and employ Nano Banana Pro~\cite{nanobanana}." The provided bibliography entry for `gemini` is titled "Gemini 3 Pro" (year 2026), and `nanobanana` is titled "Gemini 2.5 Flash Image (Nano Banana)" (year 2025). The text claims the existence of a "3.1" version and a "Pro" variant of Nano Banana which are not reflected in the citation keys or titles. While these may be internal codenames or very recent updates, the current citations do not support the specific version numbers "3.1" and "Pro" for Nano Banana. This creates a minor factual gap where the text asserts a specific model version that the bibliography does not explicitly validate.

Second, the claim in the Introduction and Table 1 that EvalVerse is the "first" to achieve full-modality coverage (specifically "Video with Sound" and "Multi-Shot") is strong. While Table 1 correctly marks existing benchmarks like VBench and EvalCrafter with '×' for these modalities, the text in the Related Work section acknowledges that "Stable Cinemetrics" has "Partial" pipeline awareness and "VADB" has "Expert-Guided" evaluation. The claim of being the "first" to cover *both* sound and multi-shot simultaneously is likely accurate based on the table, but the phrasing should be carefully checked to ensure it doesn't inadvertently dismiss the partial capabilities of the cited works in a way that contradicts the "Partial" or "High" ratings given to them in the tables. The current text is mostly consistent, but the absolute "first" claim relies heavily on the specific combination of modalities which should be explicitly highlighted as the differentiator to avoid overstatement.

Finally, in Section 6, the use of "DINO" is cited as `oquab2023dinov2`. The paper should explicitly state "DINOv2" in the text to match the citation, as the original DINO (Caron et al.) and DINOv2 (Oquab et al.) are distinct models with different capabilities. If the evaluation pipeline relies on the specific features of DINOv2, the text should reflect that precision to ensure the citation accurately supports the claim.

Action items.

- **[writing]** In Section 5, the text cites 'Gemini 3.1 Pro' and 'Nano Banana Pro', but the bibliography only lists 'Gemini 3 Pro' and 'Nano Banana' (Gemini 2.5 Flash Image). The specific version '3.1' and the 'Pro' suffix for Nano Banana are unsupported by the provided citations. Verify model versions or update citations.
- **[writing]** Table 1 claims EvalVerse is the 'first' to cover 'Video with Sound' and 'Multi-

Shot'. While the table shows others lack these, the text must ensure this 'first' claim doesn't contradict the 'Partial' or 'High' ratings given to VADB/Stable Cinemetrics in other dimensions. Clarify that the 'first' claim applies specifically to the simultaneous combination of these modalities.

- [**writing**] In Section 6, the text cites 'DINO' using the key 'oquab2023dinov2'. Since this key refers to DINOv2, the text should explicitly state 'DINOv2' to ensure the citation accurately supports the specific model capabilities used in the pipeline.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear and comprehensive visual overview of the five-step EvalVerse framework with well-defined boxes and flows. However, the provided caption text is incomplete and cut off, failing to fully describe the final steps of the pipeline.

Figure 2

Figure 2 is a clear and well-organized taxonomy diagram that effectively mirrors the professional cinematic workflow as described in the caption. The hierarchical structure, from production stages to granular rationales, is visually distinct and easy to follow without clutter or missing labels.

Figure 3

Figure 3 effectively visualizes the annotation, sampling, and test pair construction pipeline. However, the 'Incomplete Case' label is ambiguous, and the sampling charts lack context regarding the total sample size.

Figure 4

Figure 4 presents a comprehensive comparison of video generation models across seven cinematic aspects, but the y-axis lacks a descriptive label for the metric shown. Additionally, the uneven number of models evaluated in 'Multi-Shot cutting' and 'Sound design' compared to other categories may mislead readers regarding the completeness of the benchmark in those specific areas.

Figure 5

Figure 5 presents a radar chart comparing model performance in T2V but lacks a visible legend, explicit axis definitions for the T2V context, and error bars, reducing its interpretability and scientific completeness.

Figure 6

Figure 6 presents a fine-grained comparison of models in the R2V setting using radar charts, but it lacks numerical axis scales, making absolute performance assessment impossible. Additionally, the legend is positioned only within the first subplot, creating a disjointed user experience for the remaining charts.

Figure 7

The figure effectively visualizes the alignment between human and machine judgments with clear scatter plots and trend lines. However, the caption contains a broken LaTeX symbol for the Pearson coefficient, and the legend text is inconsistently formatted.

Action items.

- [**writing**] Figure 1: The caption is truncated mid-sentence at 'human aesthetic perce' and ends with a file reference '[teaser_new.pdf]' instead of completing the description of the Expert-Machine Calibration step.
- [**writing**] Figure 3: The vertical label 'Incomplete Case (Preview)' on the left panel is ambiguous; it is unclear if it refers to the specific JSON example shown or the entire annotation pipeline.
- [**science**] Figure 3: The 'Sampling' section displays multiple donut charts with percentage values, but lacks a legend or title specifying the total sample size (N) or the specific dataset subset these distributions represent.
- [**science**] Figure 4: The 'Multi-Shot cutting' and 'Sound design' categories show sparse data with only 2-3 models evaluated, while others have 10+, making direct performance comparisons across categories misleading without noting the different evaluation scopes.
- [**writing**] Figure 4: The y-axis label is missing; while gridlines and values (0.000000 to 0.900000) are present, the metric being measured (e.g., 'Win Rate', 'Score') is not explicitly labeled on the axis.
- [**writing**] Figure 4: The legend at the top is extremely dense with 11 entries, making it difficult to distinguish between similar colors (e.g., the various shades of blue and green) without careful cross-referencing.
- [**science**] Figure 5: The radar chart axes (e.g., 'Action: Action-Emotion Synergy', 'Visual Quality: Rendering Quality') are not explicitly defined as T2V-specific metrics in the caption, creating ambiguity about whether these dimensions apply to the T2V setting or are generic across all settings.

- **[writing]** Figure 5: The legend is missing from the rendered image; while model names are listed in the caption, the specific color and line-style mappings (e.g., Hailuo 2.3 vs. Wan 2.2) are not visually labeled on the figure itself, making it difficult to distinguish models without external reference.
- **[science]** Figure 5: No error bars or confidence intervals are shown on the radar chart data points, despite the caption implying a 'fine-grained performance comparison' which typically requires uncertainty quantification for scientific rigor.
- **[writing]** Figure 6: The legend is located in the bottom-left corner of the 'Visual Concept Design' subplot, which is visually disconnected from the other six subplots, making it difficult to associate the model keys with the data in the 'Acting', 'Aesthetics', and other charts.
- **[science]** Figure 6: The radar charts lack a visible numerical scale or axis ticks (e.g., 0, 2, 4, 6, 8, 10). Without these reference points, it is impossible to determine the absolute performance scores of the models, rendering the comparison purely relative and subjective.
- **[writing]** Figure 7: The caption contains a LaTeX artifact ('$$') where the Pearson correlation coefficient symbol (ρ) should be, rendering the text 'Pearson's $$' incomplete.
- **[writing]** Figure 7: The legend labels are cluttered and inconsistent; some entries include 'win ratio' (e.g., 'Hailuo 2.3 win ratio') while others do not (e.g., 'LTX2 win ratio'), creating visual noise.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a significant density of specialized jargon and acronyms that hinders accessibility for non-specialist readers, particularly in the abstract and introduction. The term "agentic workflows" (Abstract, Introduction) is non-standard; "agent-based workflows" or "workflows utilizing autonomous agents" would be clearer. The acronym "CoT" (Chain-of-Thought) is used frequently but lacks a clear, explicit definition at its very first occurrence in the Abstract (it appears as "Chain-of-Thought reasoning" but the acronym is not formally introduced as "CoT (Chain-of-Thought)" until later or implied). The phrase "pipeline-aware" (Abstract, Section 1) is used as a compound adjective without definition; "aligned with the filmmaking pipeline" is more transparent. Similarly, "digitization of subjective cinematic expertise" (Abstract) uses "digitization" metaphorically in a way that could be simplified to "converting... into digital metrics." In Section 4.1.1, "perception prior" is technical jargon that should be phrased as "perceptual priors" or "prior knowledge from perception." Section 5.1 introduces "Out-of-Domain (OOD) failures" without fully spelling out "Out-of-Domain" at first use in the main text, assuming reader familiarity. Finally, "reward hacking" in the Conclusion is a specific term of art that requires a brief explanatory phrase for general understanding. These instances collectively create a barrier to entry for readers outside the immediate sub-field of AI evaluation.

Action items.

- [**writing**] Replace ‘agentic workflows’ with ‘agent-based workflows’ or ‘workflows using autonomous agents’ in the Abstract and Introduction. ‘Agentic’ is non-standard jargon that obscures meaning for general readers.
- [**writing**] Define ‘CoT’ (Chain-of-Thought) at its first occurrence in the Abstract. Currently, it appears as ‘Chain-of-Thought reasoning’ but the acronym ‘CoT’ is used immediately after without explicit definition in the abstract text itself, and later in the body without a clear ‘CoT (Chain-of-Thought)’ definition on first use in Section 1.
- [**writing**] Replace ‘pipeline-aware’ with ‘aware of the production pipeline’ or ‘aligned with the filmmaking pipeline’ in the Abstract and Section 1. ‘Pipeline-aware’ is a compound adjective that functions as jargon without clear definition for non-specialists.
- [**writing**] Replace ‘digitization of subjective cinematic expertise’ with ‘converting subjective cinematic expertise into digital metrics’ in the Abstract. ‘Digitization’ in this context is metaphorical jargon that could be clearer.
- [**writing**] Replace ‘perception prior’ with ‘prior knowledge from perception’ or ‘perceptual priors’ in Section 4.1.1. ‘Perception prior’ is technical jargon that may confuse readers unfamiliar with Bayesian terminology in this specific context.
- [**writing**] Replace ‘Out-of-Domain (OOD) failures’ with ‘failures when encountering data outside the training distribution’ in Section 5.1. While OOD is common in ML, it should be spelled out fully at first use in the main text for broader accessibility.
- [**writing**] Replace ‘reward hacking’ with ‘exploiting the reward function to achieve high scores without genuine quality’ in the Conclusion. ‘Reward hacking’ is a specific term of art that should be briefly explained for non-specialist readers.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent logical framework for shifting video evaluation from “rightness” to “goodness” via a pipeline-aware taxonomy and expert-calibrated VLMs. The causal chain from identifying the gap (lack of cinematic expertise in metrics) to the solution (taxonomy + CoT + SFT) is generally sound. However, there are specific logical inconsistencies in the interpretation of results and the definition of mechanisms that require clarification.

First, in Section 5.2 (“Alignment Analysis”), the authors argue that abstract dimensions (Multi-Shot, Sound) achieve the “highest agreement” with human experts because they are calibrated by task-specific SFT. While Table 2 (tab:correlation) shows high Spearman coefficients for these dimensions (e.g., Vocal SRCC = 0.9487), the "Model Number" column indicates these results are derived from only 4 or 5 models, compared to 11 for pixel-grounded dimensions. Logically, a correlation coefficient derived from a sample size of 4-5 is statistically fragile and prone to overfitting or chance alignment. The conclusion that SFT is the "decisive step" for these dimensions is not fully supported by the data presented, as the small sample size prevents a robust comparison of variance. The claim of "highest agreement" is logically weak without addressing the statistical power of the small N.

Second, the "Context-Aware Gating" mechanism described in Section 4.2 contains a circularity in its logical definition. The mechanism is defined as an indicator function $\mathbb{I}_{gate}(p, C)$ that bypasses specific metrics if the narrative context C does not warrant them. However, the paper does not explain how the VLM derives C (the narrative context) independently. If C is derived from the prompt p or the video V, the VLM must already possess the capability to understand the context to decide whether to evaluate it. If the VLM cannot understand the context, it cannot gate the metric; if it can, the gating is redundant or the definition of "context" is ambiguous. The logical flow of how the system determines "necessity" without already performing the evaluation is missing.

Finally, the claim in Section 3 that the "Real-to-Gen" data engine eliminates "stochastic bias" is logically inconsistent with the methodology. The engine uses Gemini 3.1 Pro to synthesize test prompts from metadata. If the LLM synthesizing the prompt introduces its own hallucinations or biases relative to the ground-truth video, the resulting test pair is not a faithful representation of the source. Therefore, the claim that the process eliminates stochastic bias is unsupported; it merely shifts the source of potential bias from the sampling method to the prompt generation model. The paper needs to logically address how the prompt synthesis is constrained to ensure it does not introduce new biases that invalidate the "Real-to-Gen" premise.

Action items.

- **[science]** In Section 5.2 (Alignment Analysis), the text claims that abstract dimensions (Multi-Shot, Sound) achieve the 'highest agreement' with humans due to SFT. However, Table 2 shows SRCC values for these dimensions (e.g., Vocal: 0.9487, Logic: 0.9000) are based on N=4 or N=5 models, whereas pixel-grounded dimensions have N=11. The small sample size for the 'highest agreement' claim creates a logical gap regarding statistical robustness and potential overfitting to specific model behaviors.
- **[science]** Section 4.2 defines the 'Context-Aware Gating' mechanism as an indicator function I_gate(p, C) that bypasses metrics if context C does not warrant them. However, the paper does not logically explain how the VLM determines 'C' (narrative context) independently of the prompt 'p' or the video content, nor how this gating is implemented without circular reasoning (i.e., the VLM must understand the context to decide whether to evaluate the context).
- **[science]** The 'Real-to-Gen' data engine (Section 3) claims to eliminate 'stochastic bias' by sampling from a professional database. However, the construction of test pairs relies on Gemini 3.1 Pro to synthesize prompts. If the prompt generation itself introduces bias or

hallucination relative to the ground truth video, the claim of eliminating stochastic bias is logically unsupported, as the evaluation input is no longer a direct reflection of the source material.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the capabilities and impact of EvalVerse that extend slightly beyond the provided empirical evidence, particularly concerning the scope of its current validation and the statistical robustness of its alignment claims.

First, the Abstract and Conclusion repeatedly assert that EvalVerse "establishes a fundamental infrastructure" and "catalyzes the transformation" of generative models into virtual directors by providing reward signals for RL and agentic workflows. While the framework is designed with this intent, the paper currently only validates the *evaluation* component (benchmarking and correlation with humans). It does not present results where EvalVerse is actually used to train a video generation model via RL or to control an agent loop. Claiming it "establishes" this infrastructure implies a functional deployment that has not yet been demonstrated. The language should be tempered to reflect that the framework *enables* or *is poised to serve as* this infrastructure, pending future integration work.

Second, the claim that the framework "successfully digitizes subjective, expert-level cinematic knowledge" (Abstract, Section 1) is not fully supported by the statistical data presented in Table 4. While many dimensions show strong correlation, the "Sound Design" and "Multi-Shot" categories—central to the paper's argument about evaluating "goodness" beyond simple visual fidelity—exhibit non-significant p-values (e.g., Vocal p=0.1540, Soundscape p=0.1498, Rhythm p=0.0820). A p-value > 0.05 indicates that the observed correlation could plausibly occur by chance given the small sample size (N=4 or N=5 models). Therefore, stating that the digitization is "successful" across *all* dimensions is an over-interpretation of the data. The authors should qualify this claim to acknowledge that alignment is robust for visual dimensions but remains statistically inconclusive for complex audio-visual and temporal dimensions.

Finally, the Introduction states that the "Real-to-Gen" data engine "eliminates the stochastic bias inherent in existing prompt-based benchmarks." This is an absolute claim that ignores the potential for selection bias within the source "million-scale professional database." If the database over-represents certain genres (e.g., action vs. drama) or eras, the benchmark inherits that bias. "Eliminates" is too strong; "mitigates" or "reduces" would be more scientifically accurate, accompanied by a brief discussion of the dataset's composition limits.

Action items.

- [**writing**] The claim that EvalVerse 'establishes a fundamental infrastructure' for RL and agentic workflows (Abstract, Conclusion) overreaches the current evidence. The paper only presents a benchmark and a fine-tuned evaluator; it does not demonstrate the framework actually training a video model via RL or driving an agent loop. Rephrase to ‘potential to serve as’ or provide a pilot RL experiment.

- [**science**] The statement that the framework ‘successfully digitizes subjective, expert-level cinematic knowledge’ (Abstract, Introduction) is an over-claim. While correlation coefficients are high (Tab. 4), the p-values for Multi-Shot and Sound Design dimensions are >0.05 (e.g., p=0.1540 for Vocal), indicating the alignment is not statistically significant for these critical ‘goodness’ dimensions. Qualify the claim of ‘successful digitization’ to reflect these statistical limitations.
- [**writing**] The assertion that the ‘Real-to-Gen’ data engine ‘eliminates the stochastic bias inherent in existing prompt-based benchmarks’ (Introduction) is too absolute. Sampling from a professional database introduces its own selection biases (e.g., genre, era, style) which are not quantified. Replace ‘eliminates’ with ‘mitigates’ and briefly acknowledge the potential for dataset-specific bias.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a sophisticated framework for evaluating professional cinematic video generation, but it raises significant safety and ethical concerns that require clarification before acceptance.

First, regarding human subjects research, Section 5.1 details a “Human Evaluation Protocol” involving a multi-disciplinary team of 34 experts who performed side-by-side comparisons and provided discriminative rankings. However, the manuscript lacks an Ethics Statement or any mention of Institutional Review Board (IRB) or equivalent approval. Given that these individuals are providing subjective judgments that form the ground truth for the benchmark, the authors must explicitly state whether this research was reviewed by an ethics committee, how informed consent was obtained, and what measures were taken to ensure the privacy and fair compensation of the annotators. Without this, the reproducibility and ethical standing of the human evaluation component are compromised.

Second, the data provenance and copyright implications are unclear. Section 4 (“Dataset Curation”) states that the benchmark is constructed from a “million-scale professional database” of films and animations. The paper does not specify the source of this database, the licensing terms under which these copyrighted works are held, or the legal basis for creating derivative “Real-to-Gen” test pairs. If the benchmark or the underlying data is released, it could expose users to significant copyright infringement risks, particularly given the current legal landscape surrounding generative AI and training data. The authors must clarify the copyright status of the source material and the licensing of the resulting benchmark dataset.

Finally, the paper explicitly positions EvalVerse as a “fundamental infrastructure” for Reinforcement Learning (RL) and agentic workflows (Abstract, Conclusion). This creates a dual-use risk: the same expert-calibrated evaluator designed to improve cinematic quality could be used to optimize models for generating harmful content, such as non-consensual deepfakes, disinformation, or content violating safety guidelines, by providing a high-fidelity reward signal for “realism” or “professionalism.” The authors should include a discussion on these potential misuse scenarios and outline any mitigation strategies, such as safety guardrails or usage policies, intended to accompany the release of the benchmark and the evaluator model.

Action items.

- [**writing**] The paper presents a sophisticated framework for evaluating professional cinematic video generation, but it raises significant safety and ethical concerns that require clarification before acceptance. First, regarding human subjects research, Section 5.1 details a “Human Evaluation Protocol” involving a multi-disciplinary team of 34 experts who performed side-by-side comparisons and provided discriminative rankings. However, the manuscript lacks an Ethics Statement or any mention of Institutiona

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a comprehensive framework for evaluating cinematic video generation, but the scientific evidence supporting the reliability of its human and machine evaluation components requires significant strengthening.

Human Evaluation Rigor: In Section 5.1 (“Human Evaluation Protocol”), the authors describe a multi-disciplinary team and a three-stage pipeline but fail to provide essential statistical metadata. The manuscript does not state the total number of human annotators, the number of samples each annotator reviewed, or the inter-annotator agreement (IAA) scores (e.g., Cohen’s Kappa or Krippendorff’s Alpha). Without IAA metrics, the claim that the human annotations constitute a reliable “ground truth” is unsubstantiated. High variance in subjective cinematic judgment is expected; the paper must demonstrate that the expert panel reached a consensus sufficient to serve as a benchmark.

Statistical Power in Alignment Analysis: The most critical scientific weakness lies in the sample sizes reported in Table 3 (“Human-machine alignment: correlation coefficients”). For the “Multi-Shot” dimensions (Logic, Rhythm) and “Sound Design” dimensions (Vocal, Soundscape), the model count (N) is listed as 5 and 4, respectively. Calculating Spearman Rank Correlation (SRCC) and Pearson Linear Correlation Coefficient (PLCC) on such a tiny sample size ($N < 10$) is statistically unsound. The reported p-values (e.g., $p=0.0513$ for Vocal SRCC) are highly unstable and likely inflated due to the lack of degrees of freedom. The authors must either expand the evaluation to include a significantly larger set of models for these specific dimensions or explicitly acknowledge that the correlation results for these categories are preliminary and not statistically significant.

Calibration Evidence: While the "Progressive Calibration Mechanism" (Sec. 6.1) is conceptually sound, the paper lacks ablation studies or quantitative evidence showing the specific contribution of each tier (Prompt-Level, Fusion-Level, Parameter-Level) to the final alignment score. The claim that SFT is the "decisive step" for abstract dimensions is asserted but not rigorously proven with controlled experiments isolating the effect of the fine-tuning versus the prompt engineering.

To meet the standards of scientific evidence, the authors must provide the missing statistical details regarding the human evaluation process and address the severe sample size limitations in the correlation analysis for complex modalities.

Action items.

- [**science**] The human evaluation protocol (Sec. 5.1) lacks critical statistical details: the exact number of expert annotators, their inter-annotator agreement (e.g., Krippendorff's alpha), and the total number of video samples evaluated per model. Without these, the robustness of the 'ground truth' cannot be assessed.
- [**science**] Table 2 (win_ratio) and Table 3 (correlation) report high alignment metrics, but the sample sizes (N) for the 'Multi-Shot' and 'Sound Design' dimensions are critically low (N=5 and N=4 respectively). These small N values render the reported p-values (e.g., p=0.0513) statistically unreliable and prone to Type I errors.
- [**science**] The paper claims a 'three-stage quality control pipeline' for human annotation but provides no quantitative evidence of its efficacy, such as the rejection rate of initial annotations or the specific agreement metrics between the 'Quality Inspectors' and 'Final Reviewers'.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis presented in the paper, particularly in Section 5.3 ("Alignment Analysis") and Tables 4 and 5, demonstrates a generally sound approach to validating the human-machine alignment of the EvalVerse framework. The use of Spearman Rank Correlation (SRCC) and Pearson Linear Correlation Coefficient (PLCC) is appropriate for assessing the monotonic and linear relationships between the automated evaluator's scores and human expert rankings. The reporting of p-values alongside correlation coefficients is a positive step toward statistical rigor.

However, several statistical concerns regarding sample size, multiple comparisons, and uncertainty quantification require attention before the claims of "robust alignment" can be fully accepted:

1. **Small Sample Sizes and Statistical Significance:** In Table 5 (`tab:correlation`), the analysis for "Multi-Shot" (Logic, Rhythm) and "Sound Design" (Vocal, Soundscape) dimensions is based on extremely small sample sizes (N=4 or N=5). For instance, the p-value for the SRCC of "Vocal" is 0.1667, and for "Soundscape" it is 0.0513. These values are not statistically significant at the conventional $\alpha = 0.05$ level. The text claims "consistently high correlation scores" and "robustly align," which is misleading for these specific dimensions where the null hypothesis of no correlation cannot be rejected. The authors must explicitly acknowledge the lack of statistical significance for these sub-dimensions due to low power, rather than grouping them with the highly significant results from other dimensions.
2. **Multiple Comparisons Problem:** The study evaluates alignment across 18 distinct sub-dimensions (Table 5). Conducting 18 separate hypothesis tests without correction inflates the family-wise error rate. Several reported p-values (e.g., 0.0513, 0.0729, 0.0820) are borderline. The authors should apply a correction method (e.g., Bonferroni or Benjamini-Hochberg) to control for false discoveries. If corrected, many of the "significant" findings for the smaller sample sizes would likely become non-significant, necessitating a more nuanced discussion of the framework's reliability in these specific areas.
3. **Lack of Uncertainty Quantification:** The win-ratio analysis in Table 4 (`tab:win_ratio`) and the discussion in Section 5.3.1 present point estimates (e.g., "0.61/0.63") without any measure of uncertainty. Given that these ratios are derived from a finite set of comparisons, they are subject to sampling variability. The authors should report 95% confidence intervals (e.g., using the Wilson score interval or bootstrapping) for these win ratios. This would allow readers to assess whether the observed differences between machine and human win ratios are statistically distinguishable or merely within the margin of error.
4. **Assumption Verification:** The use of Pearson correlation (PLCC) assumes that the data is approximately normally distributed and that the relationship is linear. With the small sample sizes observed in the "Multi-Shot" and "Sound" categories, verifying these assumptions is difficult. The authors should explicitly state whether they verified these assumptions or if they relied solely on the robustness of the non-parametric SRCC, which is generally preferred for small, non-normal datasets.

Addressing these points will strengthen the statistical validity of the paper's central claim regarding the efficacy of the expert-calibrated evaluation pipeline.

Action items.

- [**science**] In Table 5 (tab:correlation), the p-values for 'Multi-Shot' and 'Sound Design' dimensions (e.g., p=0.0513, 0.0886) exceed the standard alpha=0.05 threshold. Given the small sample sizes (N=4 or 5), the authors must explicitly discuss the lack of statistical significance for these specific dimensions or apply a correction for multiple comparisons across the 18+ tested dimensions to avoid Type I errors.
- [**science**] The win-ratio analysis in Table 4 (tab:win_ratio) and Section 5.3.1 relies on pairwise

comparisons without reporting confidence intervals or standard errors. For a benchmark claiming ‘strong alignment,’ the authors should provide 95% confidence intervals for the reported win ratios to quantify the uncertainty of the machine-human agreement.

- [**science**] The paper reports high correlation coefficients (SRCC/PLCC) but does not specify the statistical test used to derive the p-values in Table 5. For small sample sizes (N < 30), the assumptions of the Pearson correlation test may be violated. The authors should confirm if non-parametric tests were used or justify the parametric approach given the sample size constraints.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication and presents a compelling narrative regarding the shift from “rightness” to “goodness” in video generation evaluation. The writing is generally clear, engaging, and effectively communicates the complex methodology of the EvalVerse framework. The authors successfully employ a structured approach to define their taxonomy and calibration mechanisms, making the paper accessible to readers familiar with the field.

However, there are several minor grammatical inconsistencies and stylistic choices that detract from the overall polish of the text. In the Abstract, the sentence “EvalVerse not only retains compatibility... but also significantly expands... and broaden the task coverage” suffers from a lack of parallel structure. The verb “broaden” should be “broadens” to agree with the singular subject “EvalVerse” and match the tense of “retains” and “expands.”

In the Introduction, specifically in the paragraph discussing existing benchmarks, the citation placement “They~\cite{cinetechbench...}” includes a tilde that is not standard for this context and creates a visual break in the sentence flow. While LaTeX handles spacing, the usage here is inconsistent with standard academic prose.

Furthermore, the tone occasionally slips into informality. In Section 6, the authors refer to their weight optimization method as a “trick.” In a formal scientific publication, terms like “method,” “strategy,” or “technique” are preferred to maintain professional rigor. Similarly, in the Conclusion, the phrase “for computer graphics community” is missing the definite article “the,” which is a basic grammatical oversight.

Finally, in Section 5, the term “expert-guided multi-questioning” is slightly vague. Clarifying this to “expert-guided multi-question prompts” or “expert-designed question sets” would improve precision without altering the meaning. Addressing these specific points will elevate the manuscript from a strong draft to a polished final publication.

Action items.

- **[writing]** In the Abstract, the phrase ‘broaden the task coverage’ breaks parallelism with the preceding verbs ‘retains’ and ‘expands’. Change to ‘broadens’ to maintain grammatical consistency.
- **[writing]** In Section 1 (Introduction), the sentence ‘They~\cite{cinetechbench,univbench,msvbench,muss} predominantly focus...’ contains a tilde before the citation command which is unnecessary and disrupts the flow. Remove the tilde or ensure it is used consistently for spacing elsewhere.
- **[writing]** In Section 5 (Machine Evaluation Suite), the phrase ‘expert-guided multi-questioning’ is slightly ambiguous. Consider rephrasing to ‘expert-guided multi-question prompts’ or ‘expert-designed multi-question sets’ for clarity.
- **[writing]** In Section 6 (Human-Machine Calibration), the phrase ‘data-driven weight optimization trick’ uses the word ‘trick’ which is informal for a scientific paper. Replace with ‘method’, ‘strategy’, or ‘approach’.
- **[writing]** In the Conclusion, the phrase ‘computer graphics community’ lacks a definite article. It should read ‘the computer graphics community’.